

Polynomials – 10

Where the roots lie: bounds, Gauss–Lucas, stability and
real-rootedness

Homework

24 сентября 2026 года

Theoretical minimum. Root bounds: all roots lie in $|z| \leq R$, where $R > 0$ solves $|a_n|R^n = \sum_{k < n} |a_k|R^k$ (Cauchy), or in $|z| \leq 2 \max_k |a_{n-k}/a_n|^{1/k}$ (Fujiwara); a dominant monomial forbids roots and forces signs (hence real roots); if all $a_k > 0$, then $\min \frac{a_k}{a_{k+1}} \leq |z| \leq \max \frac{a_k}{a_{k+1}}$ (Eneström–Kakeya). Gauss–Lucas, and the sign of Re or Im of $P'/P = \sum \frac{1}{z-z_j}$ off a half-plane containing the roots. Hurwitz stable polynomials have positive coefficients and are closed under products, $f \mapsto f'$, $f \mapsto f + af'$ ($a \geq 0$); Routh–Hurwitz: $a_1a_2 > a_0a_3$ for $n = 3$, $a_1a_2a_3 > a_0a_3^2 + a_4a_1^2$ for $n = 4$. Real-rootedness is preserved by $f \mapsto f + af'$ ($a \in \mathbb{R}$), by $\varphi(D)$ for real-rooted φ (Hermite–Poulain), by $\sum c_k x^k \mapsto \sum \varphi(k) c_k x^k$ for real-rooted φ without roots in $(0, n)$ (Laguerre); all $\lambda f + \mu g$ are real-rooted iff f, g interlace (Hermite–Kakeya–Obreschkoff). Descartes' rule for exponential sums $\sum c_i e^{\lambda_i x}$.

1. a) Prove Theorem 3a): a real stable polynomial has non-zero coefficients of one sign, products of stable polynomials are stable, and f' and $f + af'$ ($a \geq 0$) are stable whenever f is.
b) Prove the Routh–Hurwitz criterion for $n = 3$: if $a_0, a_1, a_2, a_3 > 0$, then $a_3z^3 + a_2z^2 + a_1z + a_0$ is stable if and only if $a_1a_2 > a_0a_3$.
c) Using the criterion for $n = 4$, determine all real t for which $z^4 + z^3 + tz^2 + z + 1$ is stable, and find its roots for the boundary value of t .
2. Prove Theorem 4b) (Hermite–Poulain) and Theorem 4c) (Laguerre), and show by the example $f = x^2 - 1$, $\varphi(t) = t - 1$ that the condition on the roots of φ in c) cannot be dropped. Deduce that for all positive

integers n and m the polynomials

$$\sum_{j=0}^n \binom{n}{j} \frac{x^j}{j!} \quad \text{and} \quad \sum_{k=0}^n \binom{n}{k} k^m x^k$$

have only real roots. (For the first one apply $(1 + D)^n$ to x^n .)

3. Let $f(x)$ be a rational function with complex coefficients whose denominator does not have multiple roots. Let $u_0, u_1, \dots, u_n$ be the complex roots of f and $w_1, w_2, \dots, w_m$ be the roots of f' . Suppose that u_0 is a simple root of f . Prove that

$$\sum_{k=1}^m \frac{1}{w_k - u_0} = 2 \sum_{k=1}^n \frac{1}{u_k - u_0}.$$

(CIIM 2011)

4. Let $f(t) = \sum_{j=1}^N a_j \sin(2\pi jt)$, where each a_j is real and a_N is not equal to 0. Let N_k denote the number of zeros (including multiplicities) of $\frac{d^k f}{dt^k}$ in the interval $[0, 1)$. Prove that

$$N_0 \leq N_1 \leq N_2 \leq \dots \quad \text{and} \quad \lim_{k \rightarrow \infty} N_k = 2N.$$

(Putnam 2000 B3)

5. Let $a, b, c, d, e > 0$ be real numbers such that $a^2 + b^2 + c^2 = d^2 + e^2$ and $a^4 + b^4 + c^4 = d^4 + e^4$. Compare the numbers $a^3 + b^3 + c^3$ and $d^3 + e^3$.

(IMC 2006)

6. Let f and g be real polynomials without common roots, with $\deg f = n \geq 1$ and $\deg g \in \{n-1, n\}$. Prove Theorem 4d): $\lambda f + \mu g$ has only real roots for all real $(\lambda, \mu) \neq (0, 0)$ if and only if f and g have only real simple roots and these roots interlace. (Consider how many times g/f takes each value $c \in \mathbb{R} \cup \{\infty\}$.)

7. a) Consider the polynomial $P(X) = X^5 \in \mathbb{R}[X]$. Show that for every $\alpha \in \mathbb{R}^*$ the polynomial $P(X + \alpha) - P(X)$ has no real roots.
b) Let $P(X) \in \mathbb{R}[X]$ be a polynomial of degree $n \geq 2$ with real and distinct roots. Show that there exists $\alpha \in \mathbb{Q}^*$ such that the

polynomial $P(X + \alpha) - P(X)$ has only real roots.

(RMO 2001)

8. Let $n \geq 2$ be a positive integer, $g \in \mathbb{R}[X]$, $g = X^n + a_1X^{n-1} + \cdots + a_{n-1}X + a_n$ a polynomial with all roots real, and $f: \mathbb{R} \rightarrow \mathbb{R}$ a polynomial function such that

$$f(x) + a_1f'(x) + a_2f''(x) + \cdots + a_nf^{(n)}(x) \geq 0 \quad \text{for all } x \in \mathbb{R}.$$

Prove that $f(x) \geq 0$ for all $x \in \mathbb{R}$.

(RMO 2025)

9. (Hermite–Biehler) Let P be a real stable polynomial of degree $n \geq 2$ with $P(0) > 0$, and write $P(z) = h(z^2) + zg(z^2)$ with real polynomials h, g .

- a) Prove that $|P(-z)| < |P(z)|$ whenever $\operatorname{Re} z > 0$, and deduce that for every λ with $|\lambda| = 1$ all roots of $P(z) + \lambda P(-z)$ lie on the imaginary axis.
- b) Deduce that all roots of h and of g are real and negative.
- c) Using that $\arg P(iy)$ is strictly increasing in y , prove that the roots of h and g are simple and interlace, the root of h closest to 0 being closer to 0 than every root of g . Recover from this the necessity of $a_1a_2 > a_0a_3$ for $n = 3$.

10. Let P be a polynomial with all real roots that satisfies the condition $P(0) > 0$. Prove that if m is a positive odd integer, then

$$\sum_{k=0}^{m-1} \frac{f^{(k)}(0)}{k!} x^k > 0$$

for all real numbers x , where $f = P^{-m}$.

(Schweitzer 1990)